Measuring Hidden-State Evolution in Chronos-2

Methods of Sequential Data

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Problem statement

A time-series foundation model processes a window of observations and forms an internal representation h_t.

$$x_{q_t-C+1:q_t} \xrightarrow{\text{frozen model}} h_t.$$

As the observation window moves, the internal representation changes.

Question: can a simple fixed model predict that change?

$$\widehat{h}_{t+k} = F_k(h_t).$$

We test the required model complexity, the useful forecast horizon and alternative explanations for predictability.

Literature overview

Representation	Snapshot / across depth	Series-time dynamics
Native, frozen	Wiliński et al. Internal Semantics Transformer Dynamics	**Our question:** predictive transitions in native TSFM states
Trained for prediction	—	LatentTSF EIDOS / LaT-PFN MIRA

Depth indexes network layers. Series-time indexes successive observation windows. We study native states without training the backbone for a new dynamics objective.

Synthetic data generation

Lorenz: convection model

$$\dot{x} = \sigma(y - x),$$
$$\dot{y} = x(\rho - z) - y,$$
$$\dot{z} = xy - \beta z.$$

$\sigma = 10$, $\rho = 28$, $\beta = 2.667$.

Rössler: chaotic oscillator

$$\dot{x} = -y - z,$$
$$\dot{y} = x + ay,$$
$$\dot{z} = b + z(x - c).$$

$a = 0.2$, $b = 0.2$, $c = 5.7$.

We observe the first coordinate x and keep the system parameters fixed across series. Initial conditions differ between series.

Source: Gilpin (2021), `dysts`, `https://arxiv.org/abs/2110.05266`.

Datasets

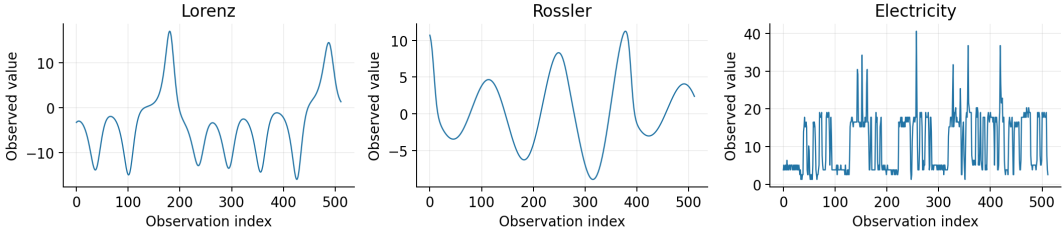


Figure 1: The first 512 observations of the first training series in each dataset. Values use each dataset's original scale.

- **Lorenz and Rössler:** observe x from the systems on the previous slide.
- **Electricity:** quarter-hour meter readings, 96 observations per day.
- **Controls:** $x_{t+1} = \phi x_t + \epsilon_t$, independent Gaussian ϵ_t. AR(1): $\phi = 0.9$; white noise: $\phi = 0$.

Electricity source: GIFT-Eval, `https://arxiv.org/abs/2410.10393`.

Hidden-state extraction

- Chronos-2: 119.5M parameters
- 12 blocks, 12 attention heads
- Hidden width 768, FFN width 3072
- Input patches: 16 observations
- Model context limit: 8192
- Our context: 512; shift: 16
- State: middle block 5 (6th of 12), last observed patch

$$h_t = E_5(x_{q_t-511:q_t}),$$

$$h_{t+1} = E_5(x_{q_t-495:q_t+16}).$$

Adjacent windows share 496 observations.

We recompute each state from its window. The encoder does not carry a recurrent state between windows.

Prediction tasks

Input: the current hidden state h_t. **Target:** h_{t+k}.

Direct prediction fits a separate map for each horizon:

$$\widehat{h}_{t+k} = F_k(h_t).$$

Recursive prediction fits one transition and applies it k times:

$$\widehat{h}_{t+1} = F(h_t), \qquad \widehat{h}_{t+k} = F^{\circ k}(h_t).$$

The main grid uses recursive prediction at $k = 1, 4, 16, 64, 256$. One base step advances the input window by 16 observations.

Transition models

For the recursive task, normalize $z = (h - \mu)/s$ and predict

$$\widehat{h}_{t+k} = \mu + s \odot F^{\circ k}(z_t).$$

$$f_\theta(z) = W_2 \tanh(W_1 z + a) + b.$$

Model	One analysis step
Ridge	$\widehat{h}_{t+1} = Ah_t + b$
MLP	$z_{t+1} = f_\theta(z_t)$
Residual	$z_{t+1} = z_t + f_\theta(z_t)$
Neural ODE	$\dot{z} = f_\theta(z)$, integrate over $\Delta t = 1$

- Neural models share a $768 \to 256 \to 768$ Tanh architecture, with separate fitted weights.
- Fit neural models on four-step segments. Select epochs using validation $k = 4$.
- Baselines: the training-target mean and persistence, $\widehat{h}_{t+k} = h_t$.

Evaluation protocol

Parameter	Value
Primary datasets	Lorenz, Rössler, electricity
Training / validation	8 / 4 separate series per dataset
Neural initialization seeds	3 for every model and dataset
Recursive horizons k	1, 4, 16, 64, 256
One analysis step	16 original observations
ODE integration	RK4, 4 substeps during training and rollout

- All models and all five horizons use the same reporting starts.
- Fit scaling and PCA on training data only.
- All reported scores use validation data. Final test partitions remain closed.

Accuracy metrics

For each validation series,

$$R^2 = 1 - \frac{\sum_t \|\widehat{h}_{t+k} - h_{t+k}\|_2^2}{\sum_t \|h_{t+k} - \bar{h}_{\mathrm{train},k}\|_2^2}.$$

- $R^2 = 0$: the training-target mean baseline
- $R^2 < 0$: worse than that baseline
- Average scores equally across validation series

We also check coordinate-standardized and whitened PCA scores. For long trajectories, we compare distributions, ACF and PSD in a training-fitted projection.

Short-horizon prediction

Dataset	Ridge $k = 1$	MLP $k = 1$	Ridge $k = 4$	MLP $k = 4$
Lorenz	0.941	0.913	0.658	0.840
Rössler	0.999	0.997	0.997	0.996
Electricity	0.509	0.441	0.418	0.316

Table 1: Validation R^2 on common starts. Every $k = 4$ forecast applies the fitted transition four times. Neural columns average three seeds.

- On Lorenz, MLP improves $k = 4$ from 0.658 to 0.840 ± 0.006 across seeds.
- Ridge is strongest on electricity under the tested budget.
- Near-perfect Rössler scores require the recurrence and timescale caveats.

Recursive prediction through 256 steps

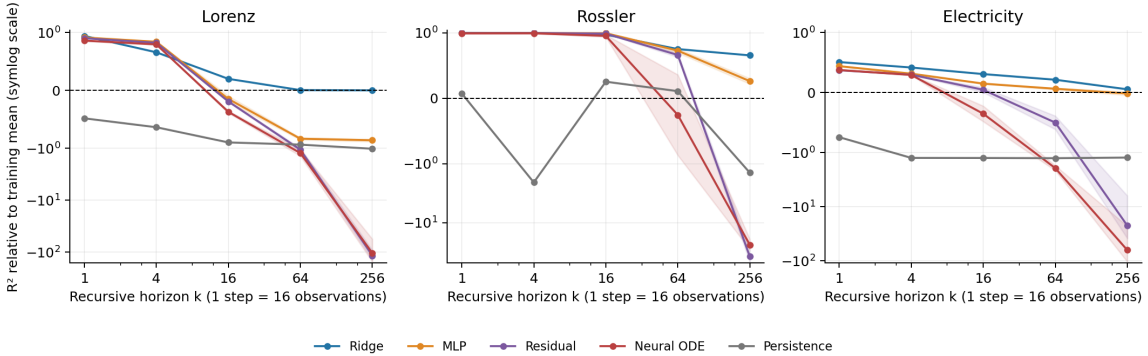


Figure 2: R^2 versus horizon k. One step is 16 observations. Neural shading shows one seed standard deviation. The vertical scale is symmetric logarithmic.

At $k = 256$, ridge retains 0.655 on Rössler, approximately zero on Lorenz and 0.055 on electricity.

Long trajectories in hidden-state space

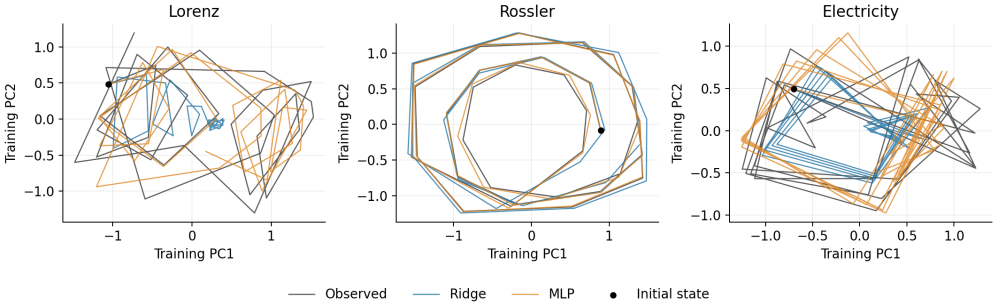


Figure 3: First 32 steps of the 256-step rollouts in training PC1–2. First fixed sampled start; MLP seed 2026. The dot marks the start. Statistics use all 256 steps.

Ridge approaches a nearly constant state on Lorenz and electricity. MLP keeps varying but loses alignment with the observed trajectory. These are diagnostic observations in a projection, not a recovered physical attractor.

Why the linear rollouts differ

For a fixed point $h^* = Ah^* + b$, a linear rollout satisfies

$$\widehat{h}_{t+k} - h^* = A^k(h_t - h^*).$$

Dataset	Spectral radius $\rho(A)$	Consequence
Lorenz	0.971	Strong asymptotic contraction
Rössler	0.99986	Oscillations decay much more slowly
Electricity	0.99634	Eventual contraction

All three radii are below one. The fixed affine maps converge asymptotically. MLP can keep varying, but its four-step training loss does not require correct phase through 256 steps.

Long-rollout metrics

For predicted and observed paths, use training PCs U and scales Λ. Let $\xi = \Lambda_8^{-1/2} U_8^\top (h - \mu)$.

$$D_{\mathrm{SW}} = \left[\frac{1}{32} \sum_{j=1}^{32} W_2^2\Big(\{u_j^\top \widehat{\xi}\}, \{u_j^\top \xi\} \Big) \right]^{1/2}.$$

$$D_{\mathrm{ACF}} = \frac{1}{8} \sum_{p=1}^{2} \sum_{\ell \in \{1,4,16,64\}} |\widehat{\rho}_p(\ell) - \rho_p(\ell)|, \qquad D_{\mathrm{PSD}} = \frac{\|\widehat{P} - P\|_F}{\|P\|_F}.$$

u_j are fixed unit directions. SW pools the last 128 states; ACF and normalized PSD use PC1–2 over the full 256-step path. Lower is better. These finite-path metrics do not prove recovery of an invariant measure.

Distribution and temporal statistics

Dataset	Model	SW	ACF error	PSD error
Lorenz	Ridge	0.989	0.071	0.621
	MLP	0.096	0.021	0.305
Rössler	Ridge	0.450	0.047	0.106
	MLP	0.046	0.016	0.085
Electricity	Ridge	0.833	0.105	0.411
	MLP	0.331	0.117	0.454

Table 2: Finite 256-step forecasts from the same 16 starts per validation series. SW: sliced Wasserstein distance in whitened PC8 over the last 128 states. ACF and PSD use PC1–2. Lower is better.

- On Lorenz and Rössler, MLP matches these statistics better than ridge, despite worse long-horizon pointwise R^2.
- A pooled distribution match alone does not establish temporal dynamics.

Forecasting the input, then encoding it

Use the observed context w_t and a predicted continuation $\widehat{v}_{t,k}$:

$$\widetilde{h}_{t+k} = E_5(\mathrm{tail}_{512}[w_t, \widehat{v}_{t,k}]) .$$

Dataset	Ridge	MLP	Chronos	Last value
Lorenz	0.612	0.810	-0.067	-0.306
Rössler	0.997	0.997	0.791	-1.048
Electricity	0.414	0.318	0.133	-0.999
White noise	0.081	—	-1.955	-1.828

Table 3: Four-step validation R^2 on a fixed subset, 32 starts per series. Chronos predicts the input median; Last value repeats the final input. Both re-encode the resulting window. Readouts use exactly the same subset.

Ridge also beats this control on white noise. A gain alone cannot establish meaningful autonomous dynamics or extra predictive information.

Neural ODE checks

- Train through RK4 with four substeps per analysis interval.
- Known stable oscillator control: four-step $R^2 > 0.999$.
- Short-horizon 8-versus-16-substep prediction difference is below 0.1% for every fit.
- Negative R^2 means larger squared error than the training-target mean.

Dataset	4 substeps	8 substeps	16 substeps
Lorenz	-102.291	-103.021	-103.054
Rössler	-22.526	-23.077	-23.056
Electricity	-61.802	-61.764	-61.764

R^2 at $k = 256$ on the same 16 starts per validation series, averaged over three seeds. This is a subset of the main table. Finer integration preserves the poor long forecast. This concerns the tested field and training setup.

Answers to the poster questions

Poster question	Evidence obtained
Can native frozen states be predicted?	Yes, at dataset-dependent horizons. Nonlinear gains are selective.
H1: smoother states and gains over persistence	Prediction gains exist. Greater smoothness is not universal and changes with scaling.
H2: temporal gains over depth imply dynamics	The threshold also passes for a constant predictor on white noise. It lacks specificity.
Can a simple ODE describe the trajectories?	The fitted field fails long-horizon fidelity checks. MLP better matches some finite-path statistics.

Final test decisions on H1/H2 remain open. Prediction accuracy alone does not prove a sufficient Markov state.

Team roles

Member	Role
Nikita Aleksandrov	Experimental design, implementation, experiments, analysis and manuscript
Ilya Lobanov	Presentation of the literature overview and research question
Namig Damirov	Presentation of the datasets and baseline methods

AI assistance included experiment configurations and launch scripts, parsers, coding/debugging, result checks, drafting/proofreading, and LaTeX/PDF builds.

Conclusions

Model	Short prediction	Long recursive prediction
Ridge	Strong baseline across datasets	Rössler $R^2 = 0.655$ at $k = 256$; contraction on Lorenz and electricity
MLP	Best Lorenz $k = 4$: 0.840 ± 0.006	Worse pointwise accuracy, better measured Lorenz/Rössler path statistics than Ridge
Residual	Lorenz $k = 4$: 0.822	Large long-horizon errors on all three datasets
Neural ODE	Lorenz $k = 4$: 0.794	Poor long forecasts persist with finer integration

Model advantages depend on the dataset, horizon and metric. All results use validation data.